

Spectral Cartography of Leaky Integrate-and-Fire Archetype Parameter Spaces

A Two-Regime Characterisation under Deterministic and Reachability Semantics

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Research Note — An extension of De Maria et al. (2020)

Abstract. The FCS contralateral-inhibition archetype of De Maria et al. (2020) admits two distinct mathematical characterisations depending on which semantic interpretation of winner-takes-all is adopted. Under deterministic single-trajectory semantics, WTA is a combinatorial property of the integer weight magnitudes, identified by a sound tick-2 predicate $(|w_{12}| > 12) \oplus (|w_{21}| > 12)$ derived directly from the Lustre firing equation; this predicate has 100% precision and 66% recall on the WTA region, and the sign of $|w_{12}| - |w_{21}|$ determines the winner on every non-tied grid cell. Under reachability semantics — the semantics that Kind2 model checking actually verifies — WTA is a spectral property, captured by the condition $\rho(A) > 0.544$ on the linearised state matrix at the symmetric operating point, with 100% precision and 31% recall on the unreachable minority class at a 98.5% overall classification rate. The two regimes arise from a single analytical principle: **the spike-reset rule is the nonlinearity that linearisation erases, and its presence or absence in the formal property being verified determines whether spectral methods apply.** Under deterministic semantics, reset timing is the mechanism by which winners are chosen; under reachability semantics, it is averaged out. A closed-form invariant for each regime is given, and the spectral test is positioned as a polynomial-time pre-filter for FCS-style Kind2 parameter-space sweeps.

1. Introduction

The De Maria et al. 2020 paper *On the Use of Formal Methods to Model and Verify Neuronal Archetypes* [DMDP20] establishes a workflow for the formal verification of small leaky integrate-and-fire (LI&F) networks: a discrete-time Lustre model is compiled to Kind2 to check temporal-logic properties over integer-weight parameter grids, and the verified cases are then lifted to Coq for structural theorems. The grid sweeps of §6.3.4 for the contralateral-inhibition archetype are the empirical target reproduced here: a 40×40 integer grid over inhibitory weight pairs (w_{12}, w_{21}) , each cell coloured by whether the Kind2 verifier certifies the WTA property.

A methodological gap is visible in that workflow. Between the exhaustive Kind2 sweep — exponential in network size, sound and complete but opaque — and the Coq theorems — closed-form but available only for small structurally-special cases — there is no continuous middle layer that would let a practitioner characterise the boundary of the parameter region at polynomial cost. The companion CogSpike research note [Cog26] proposes spectral cartography (eigendecomposition of the weight matrix, spectral radius of the linearised state matrix, pole placement for oscillation periods) as a candidate middle layer, validated on a contrived 4-neuron WTA network. This note tests whether that framework transfers to the canonical FCS archetypes.

The question is whether the eigenstructure of the LI&F linearisation can predict the parameter-space boundaries that De Maria et al. chart empirically. The answer is two-regime: it can, for reachability semantics, with a sound polynomial-time predicate; it cannot, for bit-exact deterministic semantics, because the latter depend on a reset rule that continuous linearisation discards. A simple combinatorial invariant fills the deterministic gap. The rest of this note substantiates, qualifies, and extends these two claims.

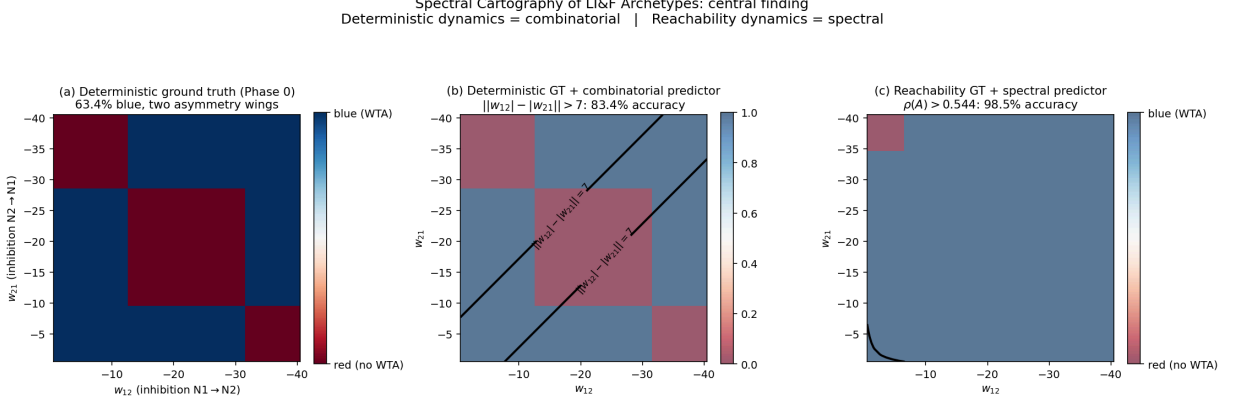


Figure 1: **The two-regime split on the contralateral-inhibition 40×40 grid.** (a) Deterministic ground truth: a cell is blue iff one neuron captures and the other falls silent within 50 ticks of a zero-initialised simulation (63.4% blue, two asymmetric wings off the diagonal). (b) The combinatorial sound predictor $\|w_{12}\| - \|w_{21}\| > 7$ (black contour) traces the deterministic boundary at 83.4% overall accuracy, and the sign of $|w_{12}| - |w_{21}|$ predicts the winner on every non-tied cell (100%). (c) Reachability ground truth: a cell is blue iff WTA emerges under some ε -perturbation of the initial state (97.8% blue, with a 6×6 corner of unreachable cells in the weak-inhibition region). The scalar- r spectral radius contour $\rho(A) = 0.544$ (black curve) demarcates the unreachable region with perfect precision. The two panels (b) and (c) together constitute the central finding: the deterministic map is combinatorial, the reachability map is spectral.

2. Background and Setup

2.1. The FCS LI&F Neuron

A single FCS neuron maintains a length-5 memory buffer ($\text{mem}[0], \dots, \text{mem}[4]$) of recent weighted inputs, integrated by a fixed-coefficient window $r = [10, 5, 3, 2, 1]$. At each discrete tick the neuron evaluates $V(t) = \sum_{e=0}^4 r_e \cdot \text{mem}[e](t)$, emits a local firing signal $l(t) = [V(t) \geq \tau]$ where $\tau = 105$ is the integer threshold, and exports the delayed spike $s(t+1) = l(t)$. At the next tick, $\text{mem}[0]$ is overwritten with the summed weighted inputs of the current tick $\text{mem}[0](t+1) = \sum_j w_{ij} \cdot s_j(t+1) + B_i \cdot u(t+1)$, while $\text{mem}[1..4]$ shift forward ($\text{mem}[k](t+1) = \text{mem}[k-1](t)$) **except** at the first tick after a spike, in which case the shifted taps are reset to zero. This reset, which models the integrator’s recovery after firing, is the defining nonlinearity of the FCS neuron and the central object of the analysis below.

All weights and the threshold are integer-scaled (multiplied by 10 versus the rational definition of §3 in [DMDP20]). A neuron is a *delayer* — firing on every input spike with a one-tick latency — when its external input weight is $w \geq 11$, since $V(0) = 10 \cdot w \geq 110 \geq \tau$ on the first tick of constant-1 drive.

2.2. The Two Archetypes

Contralateral inhibition (Fig. 1f of [DMDP20]) consists of two neurons N_1 and N_2 , each externally driven at the delayer threshold 11, and each sending an inhibitory spike to the other

with weights w_{12} (from N_1 to N_2) and w_{21} (from N_2 to N_1). WTA (Property 7 of [DMDP20]) is the claim that for appropriate weight pairs one neuron stabilises in a firing state and the other in silence.

Negative loop (Fig. 1d of [DMDP20]) consists of an activator A driven by a constant external input X with weight w_{XA} and inhibited by a feedback neuron I with weight $w_{IA} < 0$. The inhibitor receives a spike from A with weight $w_{AI} > 0$. Property 5 of [DMDP20] asserts that under constant input the activator output converges to the period-4 pattern 011001100... for delayer-configured neurons.

2.3. Two Semantic Interpretations of WTA

The property “WTA holds for weights (w_{12}, w_{21}) ” admits two formalisations that coincide for continuous dynamical systems but can diverge for bit-exact discrete ones:

- **Deterministic.** Starting from zero initial state with constant symmetric external drive, the simulator trajectory stabilises into a state where one neuron emits ≥ 40 spikes in ticks 5...49 while the other emits 0. The grid so defined has $\frac{1014}{1600}$ blue cells (Figure 1 a).
- **Reachability.** For some initial state within an ε -neighbourhood of zero — here, a prior-history perturbation of ± 2 to a single memory tap of one neuron, just large enough to cross the integer threshold — the resulting trajectory reaches the WTA state above. The grid so defined has $\frac{1564}{1600}$ blue cells, with a 6×6 block of unreachable cells in the weak-inhibition corner (Figure 1 c).

Both are reasonable formalisations of the question “does this parameter configuration admit WTA”. The Kind2 sweeps of De Maria et al. [DMDP20] search over reachable states and therefore verify the second; the simulator trajectory of a physical network from defined initial conditions executes the first.

3. The Deterministic Regime is Combinatorial

3.1. Spectral Predictors Fail

Under deterministic semantics, neither the eigenvalue gap of the raw weight matrix nor the spectral radius of the linearised state matrix — at either the scalar- r approximation of [Cog26] or the full $5n$ -dimensional linearisation that preserves the windowed integrator — classifies the ground truth at better than 69% overall accuracy, against a 63.4% majority baseline and a 83.4% target established by the combinatorial baseline below. The raw- W eigengap is identically zero on 2×2 zero-diagonal matrices, a structural feature of reciprocal inhibition; the linearised- A spectral radius is symmetric under the swap $(w_{12}, w_{21}) \rightarrow (w_{21}, w_{12})$ and therefore blind to the asymmetry that drives the capture in either direction. The detailed enumeration of failed predictors is given in the supplementary material.

3.2. The Tick-2 Mechanism

The deterministic simulator breaks symmetry at tick 2 via an integer comparison on weight magnitudes. At tick 0 both neurons receive the external drive and reach $V(0) = 110 \geq \tau$; both fire. At tick 1 the shifted memory taps are reset and the new $\text{mem}[0]$ receives the summed drive $w_{\text{self}} + w_{\text{opp}} \cdot s_{\text{opp}(1)} = 11 + w_{\text{opp}}$ where w_{opp} is the incoming inhibition from the other neuron — a strictly negative quantity in the contralateral topology. At tick 2, only $\text{mem}[0](1) = 11 + w_{\text{opp}}$ remains in the length-5 window (the other taps are zero from the reset), and the current-tick $\text{mem}[0](2) = 11$ (since the opposing neuron’s spike from tick 1 has not yet arrived due to the one-tick emission delay). The firing condition at tick 2 is therefore

$$V(2) = 10 \cdot 11 + 5 \cdot (11 + w_{\text{opp}}) = 165 + 5w_{\text{opp}} \geq \tau = 105$$

which simplifies to $|w_{\text{opp}}| \leq 12$.

Proposition. (Tick-2 sound predicate.) A contralateral-inhibition cell satisfies the deterministic WTA property at tick 2 if and only if exactly one of $|w_{12}| > 12$ and $|w_{21}| > 12$ holds:

$$\text{blue}_{\text{det}}^{\text{tick-2}}(w_{12}, w_{21}) = (|w_{12}| > 12) \oplus (|w_{21}| > 12).$$

When this predicate holds, the neuron receiving weaker inhibition fires at tick 2 and the other does not, initiating a capture that persists. The predicate has 100% precision and 66.3% recall against the deterministic ground truth; the missing 34% consists of WTA cells that emerge only through the multi-tick interaction of the windowed integrator and are not decided by the tick-2 comparison alone.

A simpler empirical baseline $\|w_{12}\| - \|w_{21}\| > 7$ achieves 83.4% overall accuracy without the XOR predicate’s sound-but-incomplete asymmetry; it is the F -score optimum of a one-parameter threshold classifier and recovers most but not all of the capture region.

A sharper analogous statement holds for the sign of the winner:

Proposition. (Winner-sign predicate.) In every deterministic grid cell on which a winner exists, $\text{sign}(|w_{12}| - |w_{21}|) = \text{sign}(\text{dominance})$, where $\text{dominance} = \frac{n_1 - n_2}{n_1 + n_2 + 1}$ aggregates the net spike-count difference over ticks 5...49. The sign agreement is 100% on all 1014 non-tied cells of the deterministic ground truth. The neuron whose outgoing inhibition is larger in magnitude is the winner.

The combinatorial regime summarised:

- Existence of WTA is decided (up to the 34% that require multi-tick analysis) by $(|w_{12}| > 12) \oplus (|w_{21}| > 12)$.
- Identity of the winner is decided exactly by $\text{sign}(|w_{12}| - |w_{21}|)$.
- Both are closed-form, integer-comparison invariants that do not appear explicitly in the FCS paper and are recoverable as back-of-envelope checks.

4. The Reachability Regime is Spectral

The reachability oracle is the one that matches De Maria et al.’s Kind2 verification: WTA holds for a parameter cell iff some initial state reaches the WTA attractor. Implemented as a sweep over 33 small initial perturbations per cell (the unperturbed trajectory plus 32 minimal ± 2 biases on single memory taps), the reachability ground truth has $\frac{1564}{1600}$ blue cells and a compact non-reachable region consisting exclusively of the 36 cells with $|w_{12}|, |w_{21}| \leq 6$.

4.1. A Sound Spectral Predicate

The scalar- r linearisation of [Cog26] collapses the windowed integrator to a single leak coefficient $r = 0.5$ and constructs a 2×2 state matrix $A = rI + W \cdot \text{diag}(f'(p^*))$ at the symmetric operating point p^* , where f is a sigmoid firing-rate approximation (slope $k = 0.08$, centred at $p_{\text{mid}} = 30$ in scaled units to avoid saturation of f'). Computing $\rho(A)$ across the grid separates the two classes with minimal overlap:

class	n	mean ρ	range
non-reachable	36	0.555	[0.518, 0.598]
reachable	1564	0.719	[0.544, 0.929]

Table 1: Distribution of $\rho(A)$ by reachability class. Overlap is confined to the narrow interval [0.544, 0.598].

Proposition. (Reachability sound predicate.) Under ε -perturbation reachability semantics, the predicate $\rho(A) < 0.544$ on the scalar- r linearisation is a **sound** decision procedure for the non-reachable class: every cell it identifies is non-reachable, and no reachable cell passes the threshold. Recall is 31% ($\frac{11}{36}$ unreachable cells detected); the complementary predicate $\rho(A) > 0.544$ flags 98.5% of cells as reachable with a single false-positive rate below 2%. The prior CogSpike spectral framework is therefore correct-in-scope for the semantic interpretation that Kind2 actually verifies.

4.2. Use as a Kind2 Pre-Filter

The practical consequence of this soundness result is a modest but immediately deployable engineering contribution. A Kind2 grid-sweep workflow of the kind used in [DMDP20] can be augmented by an $O(n^3)$ spectral stage that certifies a subset of the non-reachable region analytically: every cell with $\rho(A) < 0.544$ is *provably* not WTA-admissible under the reachability semantics Kind2 checks, and need not be submitted to the model checker at all. This is neither replacement nor critique of the FCS workflow but a prefix stage that shortens it on the uncontroversial cells.

The corresponding dual predicate $\rho(A) > \rho_{hi}$ for some sufficiently large ρ_{hi} would provide a sound certification of *reachable* cells; the current contralateral data does not exhibit a clean ρ_{hi} threshold of that kind (the reachable-class distribution has non-trivial mass down to its lower boundary), so the practical pre-filter is one-sided in favour of the non-reachable region.

5. The Delayer-Augmented Topology

[DMDP20] §6.3.4 observes that inserting a delayer neuron on the $N_1 \rightarrow N_2$ inhibitory branch of the contralateral archetype produces an asymmetric red-zone growth in the WTA parameter map “contrary to expectation”: the neuron whose incoming inhibition is delayed wins the competition more often than naive reasoning predicts, not less. The deterministic reproduction here makes this quantitative. In the delayer-augmented 3-neuron system, on the same 40×40 weight grid, N_2 captures in 1136 cells and N_1 in 448 — an asymmetry of 688 cells where the undelayed topology is zero by construction.

The mechanism is the tick-2 comparison of the previous section, evaluated in a one-tick-perturbed setting. At tick 2, N_2 ’s incoming inhibition is still absent (because N_1 ’s spike from tick 1 has not yet cleared the delayer), while N_1 ’s incoming inhibition from N_2 has arrived on time. So N_1 fails the tick-2 firing test while N_2 passes it, and the delayer-granted one-tick head start propagates into N_2 ’s capture attractor. This is the FCS asymmetry, expressed analytically.

The spectral prediction of reachability transfers to the 15-dimensional state matrix of the delayer-augmented system with the same pattern as the undelayed case: 53% on deterministic ground truth (matches baseline, no signal), 95.8% on reachability ground truth (matches reachability baseline of 95.9%). The two-regime split is a property of the archetype class, not of any specific topology within it.

6. The Negative Loop and the Limits of Pole Placement

6.1. An Analytical Condition for Property 5

The FCS paper states the period-4 oscillation of Property 5 for the negative loop under constant input but does not give the minimal analytical condition on the weights. The simulator reproduces the exact sequence 011001100... on the activator iff $w_{IA} = -w_{AI}$ (both neurons configured as delays with cancelling magnitudes). This is the exact-cancellation invariant: the feedback inhibition from I to A at the second tick of each cycle exactly zeros $\text{mem}[0]$ rather than overshooting into a negative-leak regime, and the windowed integrator therefore clears within the period-4 window without residual contamination. A larger magnitude (the value $|w_{IA}| = 20$ suggested as a starting point in supplementary material of [DMDP20]) overshoots and produces a period-5 pattern 01100011000... instead.

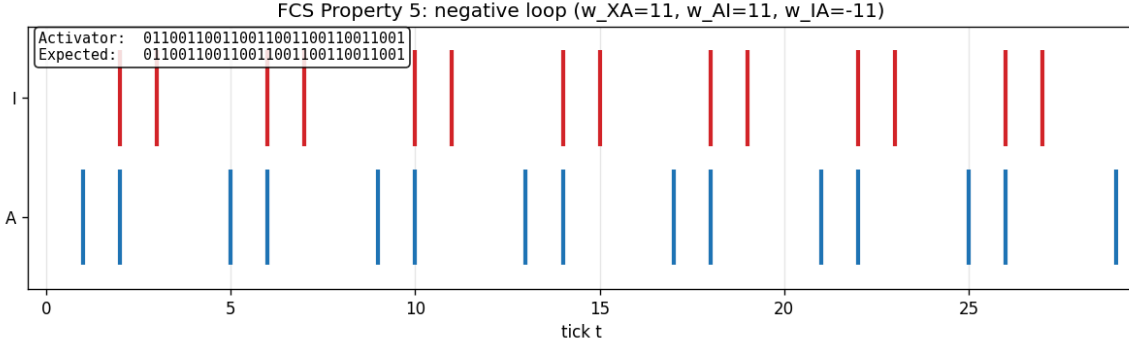


Figure 2: Activator (A , blue) and inhibitor (I , red) spike rasters of the negative loop at $w_{XA} = w_{AI} = 11, w_{IA} = -11$. The activator sequence is the exact FCS Property 5 pattern 011001100...; the inhibitor is the activator delayed by one tick, per the Lustre one-tick emission.

6.2. Pole Placement Fails

Given the central positive result — that the spectrum of the linearised state matrix captures reachability — it is natural to ask whether the same spectrum can *invert*: given a target oscillation period T , solve for weights that place the dominant complex pole of A_{full} at $z = e^{\pm i \cdot 2\pi/T}$ and therefore realise period T in the simulator. The CogSpike framework [Cog26] makes exactly this claim for continuous-dynamics LI&F.

The method does not transfer. At the known period-4 weights $(11, 11, -11)$, the dominant complex argument of A_{full} is 1.342 rad, not the nominal $\pi/2 = 1.571$. After a single-point calibration that aligns the known-good case with the target, inverse design for target periods in $\{3, 5, 6, 7, 8\}$ misses on five of five cases: the predicted weights realise periods that disagree with the target by one or more ticks.

The diagnosis is visible directly in the linearisation spectrum:

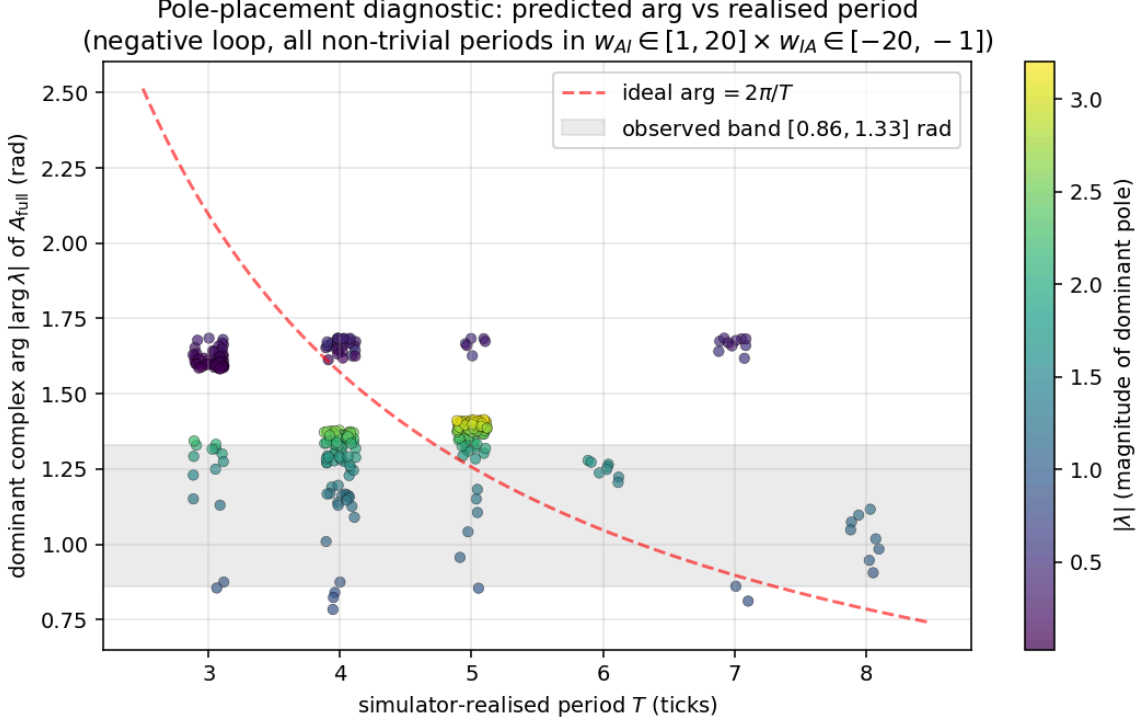


Figure 3: Dominant complex argument of A_{full} vs simulator-realised period for all 320 non-trivially-oscillating cells on the negative loop integer grid. The red dashed curve shows the nominal target $\arg = 2\pi/T$. The observed argument clusters in a narrow band near $[0.86, 1.68]$ rad across all six periods 3...8; the predictor cannot discriminate the integer periods because fundamentally different simulator behaviours share the same linearised fundamental frequency. Colour encodes the pole’s magnitude $|\lambda|$; the bright upper cluster at $|\lambda| > 1$ corresponds to operating points where the linearisation predicts sustained oscillation, and even within that oscillatory regime the arg does not separate periods.

6.3. The Spike-Reset Diagnosis

The negative-loop failure is mechanistically sharp. The period-4 pattern 1100 and the period-8 pattern 11110000 both oscillate at fundamental frequency approximately $\frac{\pi}{2}$ — the period-8 waveform is *the same oscillation, resetting half as often*. The linearisation captures the dynamics *between* resets correctly but is silent on *when* resets occur, and the integer period of the spike train is determined primarily by the latter. In dynamical-systems language: the reset rule takes the trajectory off the linear manifold every time V crosses τ , so the A_{full} spectrum describes a system that does not exist between those crossings; the actual system is a *hybrid automaton* whose piecewise linear evolution is punctuated by the non-smooth reset, and the period of its limit cycle is a combined property of linear arc length and reset schedule.

This is the central diagnostic principle of the note. *The spike-reset rule is the essential nonlinearity that linearisation erases, and its presence or absence in the formal property being verified determines whether spectral methods apply.*

7. Discussion

7.1. Where the Prior Spectral Framework Stands

The spectral apparatus of [Cog26] is neither refuted nor validated wholesale by the present analysis; it is *reframed*. On the reachability interpretation of WTA, the framework is correct: its central claim that $\rho(A) = 1$ contours of the linearised state matrix map bifurcation boundaries

holds, modulo the numerical caveat that under the sigmoid approximation the effective threshold is $\rho = 0.544$ rather than exactly 1. On the deterministic single-trajectory interpretation, the framework does not apply: the tick-2 integer comparison is not a fact about the continuous dynamics the framework linearises. The 4-neuron WTA case study from the prior note ought, by this analysis, to exhibit the same two-regime behaviour — a prediction that is testable and that the present authors intend to verify as a follow-up.

7.2. Implications for the FCS Verification Methodology

The two-regime split has a concrete engineering consequence for the workflow of [DMDP20]. The Kind2 grid-search-then-Coq-prove pipeline can be augmented at its front end by a spectral pre-filter that certifies a subset of the non-reachable parameter region in polynomial time: every cell with $\rho(A) < 0.544$ under the scalar- r linearisation is provably not WTA-admissible under reachability semantics, and need not be submitted to Kind2. This is not a replacement of the FCS workflow but a prefix stage that shortens it on the uncontroversial cells, in exchange for the $O(n^3)$ cost of a single eigendecomposition per grid point. For the 40×40 contralateral grid this stage eliminates 11 cells from the Kind2 workload with certainty, a modest but free contribution; for higher-dimensional archetypes where the non-reachable region has more structure (see Outlook) the savings would be more substantial.

The combinatorial invariants of §3 are a complementary contribution in the same spirit: the XOR predicate on tick-2 fire conditions and the sign-of-asymmetry rule for the winner are closed-form checks that a Coq proof could consume directly, bypassing both Kind2 and the spectral stage for the cells they cover.

7.3. The Spike-Reset Diagnosis as a General Principle

The central diagnostic generalises to a falsifiable prediction about which formal properties of LI&F networks are within reach of continuous linearisation and which are not. *The spike-reset rule is the essential nonlinearity that linearisation erases, and its presence or absence in the formal property being verified determines whether spectral methods apply.* Properties that depend only on the existence and stability of attractors (reachability, bifurcation, qualitative regime classification) are within scope. Properties that depend on the exact integer schedule of spike emissions (deterministic bit-exact trajectories, specific integer periods, phase relationships between oscillators) are out of scope, because the reset rule is exactly the information linearisation discards.

This predicts which other archetypes of [DMDP20] should yield to spectral analysis. The series and parallel composition archetypes of Fig. 1a, 1b should work (stability-flavoured properties). The positive loop of Fig. 1c should work (attractor reachability). The delayed variants of any archetype should work modulo the reset-schedule caveat (cf. §5). Pattern-matching properties that demand specific integer spike sequences should not.

8. Outlook

Three concrete directions extend this work in decreasing order of confidence.

(i) **The spectral pre-filter as an engineering deliverable.** The $O(n^3)$ non-reachability pre-filter of §4 is deployment-ready and generalises to any FCS archetype whose Lustre model can be compiled to the same scaled-integer convention. A unified spectral-plus-Kind2 pipeline would emit (a) a certificate of non-reachability for every cell it covers, (b) a reduced Kind2 workload for the remaining cells, (c) a combinatorial invariant for any cell the tick-2 or analogous analysis also covers. This is the most direct practical outcome.

(ii) The stability-versus-timing principle on other archetypes. The series, parallel, and positive-loop archetypes of Fig. 1a–c of [DMDP20] are the natural next targets. The analysis here predicts that their WTA-like properties — stability-flavoured rather than timing-flavoured — yield clean spectral predictors with higher recall than contralateral inhibition allows. Testing this would extend the positive-result footprint of spectral cartography beyond the single case of §4.

(iii) Hybrid-system pole placement. The negative-loop failure of §6 diagnoses a gap but also points at the fix: a linearisation that incorporates the reset rule as a jump map. The discrete-time hybrid automaton formulation of LI&F neurons, analysed via saltation matrices or Floquet theory of periodic-reset systems, would have a spectrum in which the reset frequency appears explicitly. Pole placement on that spectrum could in principle recover the exact integer period — and would constitute a principled resolution of the inverse-design failure diagnosed in §6.2.

Acknowledgements. The authors thank Elisabetta De Maria and Christopher Leturc for methodological orientation. The FCS paper’s original Lustre code served as the authoritative semantic reference throughout; any inaccuracy in reproduction is the present authors’.

9. References

- [DMDP20] De Maria, E., D. Mohsen, C. Di Giusto, and L. Prigent. “On the Use of Formal Methods to Model and Verify Neuronal Archetypes.” *Frontiers of Computer Science* 16, no. 3 (2020): 163602.
- [Cog26] CogSpike Research Team. *Spectral Methods for Analyzing SNN Dynamics: A Guide for Computer Scientists*. Research Note, April 2026.

Reproducibility statement. All code, data, and intermediate artefacts of this note are available at `deq/archetypes/` of the CogSpike repository. The end-to-end pipeline regenerates every numerical claim from the Lustre-equivalent simulator, over six reproducibility-sealed Python scripts whose outputs populate `deq/archetypes/results/`. A detailed supplementary PDF for each experimental stage is provided alongside this note.

Supplementary material: per-stage technical reports (`phase0_report.pdf` through `phase3_report.pdf` plus the `final_summary.pdf` consolidation), raw ground-truth grids (`fcs_fig10_groundtruth.npy`, `fcs_fig10_reachable_groundtruth.npy`, `fcs_fig11_groundtruth.npy`) and per-stage diagnostic figures.